



Northwind Data

BY ADELE LERMAN

A MySQL DATA ANALYSIS

Investigate product popularity: This assignment would involve using SQL queries to investigate product popularity in the "OrderDetails" table.

Calculate the total revenue generated by each product,

```
SELECT Product,  
       sum(CAST(REPLACE(`Unit  
Price`, '$', '') AS  
DECIMAL(10, 2))*Quantity)  
       as revenue  
FROM `order details`  
group by Product;
```

Product	revenue
Northwind Traders Beer	6818.00
Northwind Traders Dried Plums	262.50
Northwind Traders Dried Pears	1200.00
Northwind Traders Dried Apples	2120.00
Northwind Traders Chai	720.00
Northwind Traders Coffee	29900.00
Northwind Traders Chocolate Biscuits Mix	782.00
Northwind Traders Chocolate	2550.00
Northwind Traders Clam Chowder	2798.50
Northwind Traders Curry Sauce	2600.00
Northwind Traders Green Tea	822.25
Northwind Traders Boysenberry Spread	2500.00
Northwind Traders Cajun Seasoning	880.00
Northwind Traders Crab Meat	2208.00
Northwind Traders Ravioli	1950.00
Northwind Traders Mozzarella	3132.00
Northwind Traders Scones	200.00
Northwind Traders Olive Oil	533.75

Create a view that shows the top 10 customers by total revenue: This view would join the "Customers", "Orders", and "OrderDetails" tables to calculate the total revenue generated by each customer. The view would then sort the customers by total revenue and display only the top 10 customers.

```
CREATE VIEW `Top 10 customers by total revenue` AS
SELECT concat(`First Name`, ' ', `Last Name`) AS
CustomerName, sum(CAST(REPLACE(`Unit Price`, '$',
'')) AS DECIMAL(10, 2)) *Quantity)
AS revenue
FROM customers
JOIN orders ON customers.ID=orders.Customer
JOIN `order details` ON orders.`order ID`=`order
details`.`Order ID`
GROUP BY CustomerName
ORDER BY revenue DESC
LIMIT 10;
```

CustomerName	revenue
Amritansh Raghav	15432.50
Francisco Pérez-Olaeta	8007.50
Elizabeth Andersen	4683.00
Christina Lee	4569.00
Sven Mortensen	3786.50
Run Liu	3625.25
Soo Jung Lee	2905.50
Thomas Axen	2550.00
Karen Toh	1505.00
Roland Wacker	1412.50

Perform aggregations: Use SQL queries to calculate summary statistics for different subsets of the data. Count the number of orders placed by customers.

```
SELECT CONCAT(`FIRST NAME`, ' ',  
`LAST NAME`) AS CUSTOMERNAME,  
COUNT(`ORDER ID`) AS `TOTAL  
ORDERS`  
FROM CUSTOMERS  
JOIN ORDERS ON  
CUSTOMERS.ID=ORDERS.CUSTOMER  
GROUP BY CUSTOMERNAME;
```

CustomerName	Total orders
Karen Toh	2
Christina Lee	4
John Edwards	2
Elizabeth Andersen	6
Soo Jung Lee	4
Thomas Axen	2
Francisco Pérez-Olaeta	6
Amritansh Raghav	4
Roland Wacker	4
Peter Krschne	2
Sven Mortensen	2
John Rodman	2
Run Liu	2
Anna Bedecs	1

Perform filtering: Use SQL commands to filter the data to focus on specific subsets of the data.

Implement one of the following: Find orders placed by a specific customer: Write a query that returns all orders placed by a specific customer, identified by their contact name.

```
SELECT concat(`First Name`, ' ', `Last  
Name`) AS CustomerName, `Order ID`  
FROM customers  
JOIN orders ON  
customers.ID=orders.Customer  
WHERE concat(`First Name`, ' ', `Last  
Name`) LIKE 'Francisco Pérez-Olaeta';
```

CustomerName	Order ID
Francisco Pérez-Olaeta	37
Francisco Pérez-Olaeta	47
Francisco Pérez-Olaeta	56
Francisco Pérez-Olaeta	64
Francisco Pérez-Olaeta	74
Francisco Pérez-Olaeta	79

Join tables: Use SQL commands to join tables containing information about customers, orders, products, and employees to gain insights into the relationships between different entities in the dataset. Implement one of the following:

Find all orders and their associated products: Write a query that joins the Orders and Order Details tables to return a list of all orders and the products associated with each order.

```
SELECT orders.`Order ID`, Product
FROM orders
JOIN `order details` ON
orders.`order ID`=`order
details`.`Order ID`
ORDER BY orders.`Order ID`;
```

Order ID	Product
30	Northwind Traders Beer
30	Northwind Traders Dried Plums
31	Northwind Traders Dried Pears
31	Northwind Traders Dried Apples
31	Northwind Traders Dried Plums
32	Northwind Traders Chai
32	Northwind Traders Coffee
33	Northwind Traders Chocolate Biscuits Mix
34	Northwind Traders Chocolate Biscuits Mix
35	Northwind Traders Chocolate
36	Northwind Traders Clam Chowder
37	Northwind Traders Curry Sauce
38	Northwind Traders Coffee
39	Northwind Traders Chocolate
40	Northwind Traders Green Tea
42	Northwind Traders Boysenberry Spread
42	Northwind Traders Cajun Seasoning
42	Northwind Traders Chocolate Biscuits Mix
43	Northwind Traders Dried Plums
43	Northwind Traders Green Tea

Use subqueries: Use SQL subqueries to gain insights into specific subsets of the data. Implement one of the following: Calculate the total sales for each product category: Write a SQL subquery to calculate the total sales for each product category.

```
select Category, sum(CAST(REPLACE(`Unit
Price`, '$', '') AS DECIMAL(10, 2))
*Quantity) as `Total Sales`
from products
JOIN `order details` ON products.`Product
Name` = `order details`.`product`
JOIN `orders` ON `order details`.`Order ID` =
orders.`order ID`
group by Category;
```

Category	Total Sales
Dried Fruit & Nuts	3782.50
Beverages	22785.50
Baked Goods & Mixes	982.00
Candy	2550.00
Soups	2798.50
Sauces	2600.00
Condiments	1380.00
Jams, Preserves	5740.00
Canned Meat	2208.00
Dairy Products	3132.00
Pasta	1950.00
Oil	533.75
Grains	280.00
Canned Fruit & Vegetables	1560.00

The End

- Adele Lerman
- alerman@saintpeters.edu
- lerman.adele@gmail.com